DSC 288R
Week 4 – Model Selection and Feature Engineering

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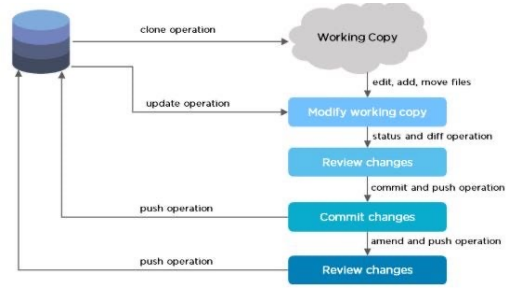
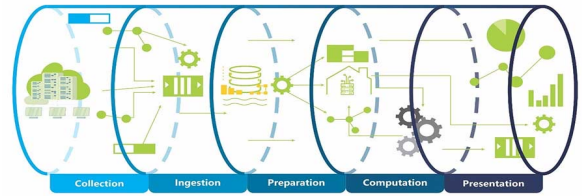
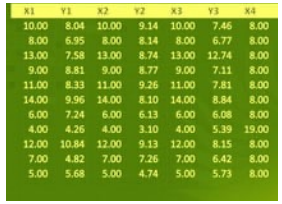
Progress so far (Week 1-3)







X1	Y1	X2	Y2	X3	Y3	X4
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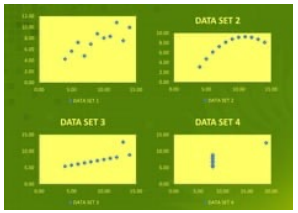
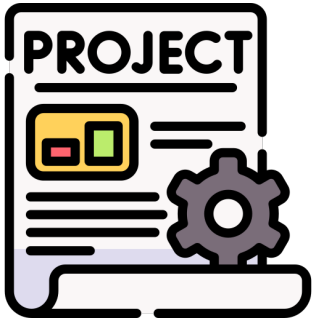
8.00	6.95	8.00	8.14	8.00	6.77	8.00
13.00	7.58	13.00	8.74	13.00	12.74	8.00
9.00	8.81	9.00	8.77	9.00	7.11	8.00
11.00	8.33	11.00	9.26	11.00	7.81	8.00
14.00	9.96	14.00	8.10	14.00	8.84	8.00
6.00	7.24	6.00	6.13	6.00	6.08	8.00
4.00	4.26	4.00	3.10	4.00	5.39	19.00
12.00	10.84	12.00	9.13	12.00	8.15	8.00
7.00	4.82	7.00	7.26	7.00	6.42	8.00
5.00	5.68	5.00	4.74	5.00	5.73	8.00



Milestones

Week	Tasks
1	Select Your Teammates
2	Submit Project Abstract
3	Create Exploratory Analysis
4	Start Modeling and Evaluation; Submit Project Progress Report
5	Continue Modeling and Evaluation
6	Continue Modeling and Evaluation
7	Prepare Project Video Presentation
8	Document Your Models
9	Submit Peer Reviews of other projects
10	Submit Final Report



Intermediate Report – Due 2/2

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1.Background - *Two lines on the problem you are planning to solve. Simply put, why is this problem important to be solved?*

2.Dataset name and citation/link - *Name of the dataset and a link to where its located.*

3.Data Pipeline

1. Purpose,
2. Pipeline design and details
3. Description of outputs (deduplicated data etc.)

4.EDA Description

1. Types of analysis used (Univariate, Multi-variate, Graphical/Non-Graphical)
2. Artifacts produced.

5.Feature Engineering

1. *Are you are doing feature engineering for one/ more of your planned models*
2. If Yes*:*
    1. *Features used in literature (if any?) that you are using*
    2. *Other features you intend using - Justify how you came up with these. Start from the EDA and trace a link to the feature.*

Intermediate Report (2)

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6. What model(s) are you using? Justify

- Name the model(s) you are using in order of highest to lowest priority for implementation

- Model Architecture from literature that you plan to use
- new architecture experiments you want to try

7.Progress Report Details on your progress.

- Effort taken on dataset collection, feature extraction, what tests you have run etc.

6.Team member contribution

- What is every member working on/planning to work on?

6.Risks and Mitigation

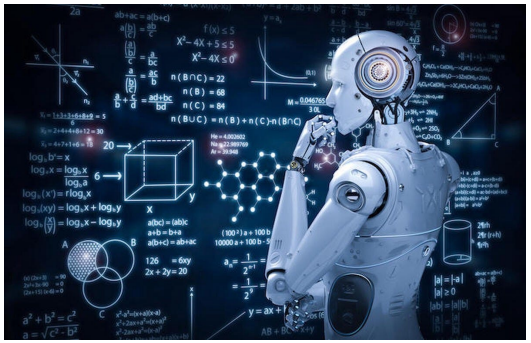
- List any risks you see at this stage to completing the project you defined in the abstract and how you are mitigating them.

6.References *- Please include relevant references.*

Model Selection

Model Selection in Data Science

- **Goal**: choose the right inductive bias for the data and task
  - Inductive bias determines what patterns a model can represent
  - Model choice impacts accuracy, robustness, and stability
  - Also affects interpretability, latency, and deployment cost
- **Focus**: principled reasoning, not model shopping



Why Model Selection Matters

- Different models fail in systematically different ways
  - Underfitting: model too simple for data structure
  - Overfitting: model memorizes noise instead of signal
  - Mismatch between data regime and model capacity
- Example: A deep NN fit training churn data well but collapsed on unseen customers.

Model Selection vs Feature Engineering

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- Feature engineering shapes the representation space
  - Determines which relationships are expressible
- Model selection shapes the hypothesis space
  - Determines how relationships are learned
- Example: Linear + interaction features rival tree models

Model Families and Core Assumptions

- **Linear models:**
  - Global additive effects
  - Limited interaction modeling
- **Tree models:**
  - Hierarchical, rule-based splits
- **Neural nets:**
  - Compositional feature hierarchies

Start with Baseline Models

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- Baselines provide a sanity check
  - Catch leakage, bugs, and label noise early
  - Establish minimum acceptable performance
  - Easier to interpret and debug

- Example: Logistic regression nearly matches XGBoost

Bias–Variance Tradeoff

- **Bias error**:
  - Model too constrained to capture structure
- **Variance error**:
  - Model overly sensitive to training data
- Model complexity controls balance
- **Example**: Polynomial regression degree tuning - Validation error clearly identified the optimal polynomial degree.

Linear Models: Strengths

- Assume linear decision boundary in feature space
  - Perform well with strong feature engineering
  - Coefficients directly interpretable
  - Fast training and inference

- Example: Credit risk models under regulation - Regulators preferred transparent coefficients over black-box gains.

Regularization in Linear Models

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- L1 regularization:
  - Encourages sparsity
  - Performs implicit feature selection
- L2 regularization:
  - Stabilizes correlated feature
- Example: Marketing attribution modeling - L1 revealed that only a few channels drove marketing impact.

Tree-Based Models

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- Automatically capture nonlinear thresholds
  - Naturally model feature interactions
  - Robust to monotonic feature transforms
- Single trees are unstable
  - Ensembles improve robustness
- Example: Fraud detection - Tree splits mirrored expert-defined fraud rules.

Bagging vs Boosting

- **Bagging**:
  - Trains models independently
  - Reduces variance via averaging
- **Boosting**:
  - Sequentially focuses on errors
  - Reduces bias

- Example: Random Forest vs XGBoost - Boosting improved recall on hard-to-detect defaulters.

GBDTs as Tabular Data Workhorse

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- Strong default for structured data
  - Handles mixed feature types
  - Captures complex interactions
  - Requires careful hyperparameter tuning

- Example: Kaggle tabular benchmarks - GBDTs consistently topped leaderboards on tabular benchmarks.

Kernel Methods (SVMs)

- Implicit feature expansion via kernels
  - Effective in high-dimensional spaces
  - Strong margins improve generalization
  - Training scales poorly

- Example: Small-sample text classification - SVMs excelled when labeled text data was scarce.

Distance-Based Models (k-NN)

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- Prediction based on similarity
  - No explicit training phase
  - Sensitive to scaling and distance metrics
  - Curse of dimensionality degrades performance

- Example: Cold-start recommendation - k-NN gave reasonable recommendations before sufficient history accumulated.

Probabilistic Models

- Explicit probabilistic assumptions
  - Model uncertainty directly
  - Often data-efficient
  - Strong independence assumptions

- Example: Naive Bayes for spam filtering - Naive Bayes remained robust despite noisy text features.

Neural Networks: Strengths

- High-capacity universal approximators
  - Learn hierarchical representations
  - Scale with data and compute
  - Require careful regularization

- Example: Image recognition with CNNs - removed the need for manual image feature design.

Deep Learning Failure Modes

- Overfitting on small or medium datasets
  - Especially severe for tabular data
- Optimization instability
  - Sensitive to initialization and learning rates
- Example: Finance datasets - Performance varied wildly across random seeds.

Model Selection for Time Series

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- Classical models:
  - Encode stationarity and seasonality
- ML models:
  - Depend on engineered lag features
- DL models:
  - Capture long-range dependencies
- Example: ARIMA vs LSTM - LSTM only helped once multiple correlated signals were added.

Evaluation Drives Selection

- Metrics encode business costs
  - Accuracy often misleading
- Cross-validation estimates robustness
  - Stratification for imbalance
- Example: PR-AUC vs ROC-AUC - ROC masked poor minority-class performance.

Operational and Ethical Constraints

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- Latency constraints limit model complexity
- Explainability may be legally required
- Fairness considerations influence model choice
- Example: Real-time credit approval - Simpler models reduced audit and compliance risk.

Deep Dive: Customer Churn

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- Dataset: 20k customers, 50 features
- Models evaluated:
  - Logistic regression (AUC ≈ 0.81)
  - XGBoost (AUC ≈ 0.83)
- Tradeoff:
  - Marginal accuracy vs interpretability
- Decision: Logistic regression deployed.
  - Business stakeholders trusted and adopted the simpler model faster.

Deep Dive: Fraud Detection

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- Dataset: 5M transactions, extreme imbalance
- Key challenges:
  - Rare positives
  - Concept drift
- Models:
  - RF vs XGBoost
- Decision: Boosted trees + PR-AUC
  - Boosted trees improved recall without overwhelming investigators.

Failure-First: When Models Break

- Deep neural networks:
  - Fail on small tabular data
- k-NN:
  - Fails in high dimensions
- SVMs:
  - Do not scale to massive datasets

- Lesson: assumptions must match data

Model Selection Decision Flowchart

- Step 1: Identify data type
  - Tabular, text, image, time series
- Step 2: Estimate data size
  - Small vs large regimes
- Step 3: Apply constraints
  - Interpretability, latency, cost

- Outcome: narrow to 2–3 candidates

Final Takeaways

- No universally best model
- Model choice is data-dependent
- Start simple and justify complexity
- Features, models, and evaluation co-evolve

- ***Successful teams revisit model choice as data evolves.***

Feature Engineering

Why Feature Engineering Dominates Model Choice

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Features define what information a model can access

No learning algorithm can recover missing information

Feature choice sets inductive bias implicitly

Bias–variance tradeoff is often controlled via representation

Feature Engineering in the ML Pipeline

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Raw data is noisy, high-entropy, and poorly structured

Logs, text, timestamps, IDs are not model-ready

Feature engineering compresses raw data into stable signals

Aggregation and normalization reduce variance

Feature Taxonomy

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Numerical features encode magnitudes and counts

Categorical features encode discrete states

Ordinal features encode ordered categories

Temporal features encode time dependencies

Text and unstructured features encode semantics

Numerical Features

- Numerical features represent measurable quantities or counts
  - They encode magnitude, frequency, or intensity
  - Examples: income, account balance, transaction count
- Example dataset: credit scoring
  - Raw inputs: annual income, total debt, monthly transactions

Numerical Features: Principles of Feature Selection

- Control scale so features interact properly with regularization
- Prefer normalized or ratio-based features
- Reduce skew and heavy tails using transforms
  - Ensure features are stable over time and populations

Numerical Features: Do's and Don'ts

- DO normalize or standardize features
- DO use log or power transforms for heavy-tailed variables
- DO clip extreme outliers deliberately
- DON'T use raw magnitudes blindly
  - DON'T let outliers dominate decision boundaries

Applying Do's and Don’ts: Credit Scoring

- Raw income → log(income)
  - Reduces dominance of very high earners
- Raw debt → debt-to-income ratio
  - Encodes repayment stress independent of wealth
- Raw transaction count → transactions per month
  - Removes exposure-time bias

Categorical Features

- Categorical features represent discrete, unordered states
  - No numeric distance exists between categories
  - Examples: country, subscription plan, device type
- Example dataset: customer churn prediction
  - Raw input: subscription plan = Basic / Plus / Premium

Categorical Features: Principles of Feature Selection

- Avoid imposing artificial ordering
- Control dimensionality as cardinality grows
- Preserve signal while avoiding label leakage
  - Plan for unseen categories at inference time

Categorical Features: Do's and Don'ts

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- DO use one-hot encoding for low cardinality
- DO use CV-safe target encoding when appropriate
- DO handle rare and unseen categories explicitly
- DON'T apply naive target encoding
  - DON'T let high-cardinality explode dimensionality

Applying Do's and Don'ts: Churn

- Subscription plan → one-hot encoding (3 plans)
  - Interpretable and low dimensional
- If plans grow → CV-safe target encoding
  - Encodes churn risk without leakage
- Unseen plan → mapped to 'other'
  - Prevents runtime failure

Ordinal Features

- Ordinal features are categorical with inherent ordering
  - Ordering matters; distances usually do not
  - Examples: credit rating, education level
- Example dataset: credit risk assessment
  - Raw input: credit rating = AAA, AA, A, BBB, ...

Ordinal Features: Principles of Feature Selection

- Preserve ordering explicitly
- Avoid encodings that destroy monotonicity
  - Check that model assumptions respect order

Ordinal Features: Do's and Don'ts

- DO encode ordinal levels as ordered integers
- DO consider monotonic constraints when available
- DON'T use one-hot encoding by default
  - DON'T assume equal spacing without justification

Applying Do's and Don'ts: Credit Rating

- AAA→1, AA→2, A→3, BBB→4, …
  - Preserves increasing risk ordering
- Tree models split naturally on thresholds
  - Linear models respect monotonic trend

Temporal Features

- Temporal features encode time-dependent structure
  - Observations are correlated across time
  - Examples: sensor readings, demand over time
- Example dataset: energy demand forecasting
  - Raw input: hourly electricity load

Temporal Features: Principles of Feature Selection

- Encode memory explicitly using lags
- Capture trends and volatility with rolling windows
- Separate calendar effects from signal
  - Respect causality strictly

Temporal Features: Do's and Don'ts

- DO use lagged and rolling features
- DO align timestamps carefully
- DO add calendar features
- DON'T use future data
  - DON'T treat time as i.i.d.

Applying Do's and Don'ts: Energy Forecasting

- Load(t−1), Load(t−24)
  - Capture short-term and daily cycles
- Rolling mean (24h)
  - Smooths transient spikes
- Hour-of-day, weekday features
  - Explain systematic demand patterns

Text Features

- Text features are unstructured and high dimensional
  - Semantic meaning is implicit, not explicit
  - Examples: reviews, support tickets, notes
- Example dataset: product review sentiment
  - Raw input: free-text customer reviews

Text Features: Principles of Feature Selection

- Reduce dimensionality while preserving meaning
- Choose representation based on task semantics
  - Balance interpretability and expressiveness

Text Features: Do's and Don'ts

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- DO start with TF–IDF for classical models
- DO use embeddings when semantics matter
- DO monitor vocabulary drift
- DON'T use massive raw bag-of-words
  - DON'T assume pretrained embeddings fit your domain

Applying Do's and Don'ts: Sentiment Analysis

- TF–IDF captures sentiment-laden terms
  - Down weights common neutral words
- Embeddings capture paraphrases
  - Generalize across phrasing
- Periodic retraining handles language drift
  - Maintains performance over time

Dimensionality Reduction

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- What it is
  - Techniques that reduce feature dimensionality while preserving signal
  - Used when features are redundant or highly correlated
- Example: numerical + interaction features in credit scoring
  - Many correlated financial ratios encode similar information
  - PCA removes correlation and stabilizes linear models
- Caution
  - Reduced features are harder to interpret and audit

Feature Leakage

- What it is
  - Using information not available at prediction time
- Example: temporal features in energy demand forecasting
  - Rolling averages accidentally include future timestamps
  - Model shows unrealistically high offline accuracy
- Why it is dangerous
  - Leaks create models that fail catastrophically in production

Missing Data as Signal

- Key idea
  - Missingness itself can carry information
- Example: credit scoring and transaction history
  - Missing income or sparse transaction data correlates with risk
  - Naive imputation destroys this signal
- Correct handling
  - Add missingness indicator features alongside imputation

Evaluating Feature Quality

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Why evaluation matters

Feature impact is often non-obvious

Example: credit scoring features

Ablation: remove debt-to-income ratio → performance drops

Permutation importance confirms dependency

SHAP values explain contribution direction

Goal

Detect spurious or leaked features early

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Thank you!